

Task 1:

Cat6 Cable T568B Wiring Standard & RJ-45 Pinout Diagram

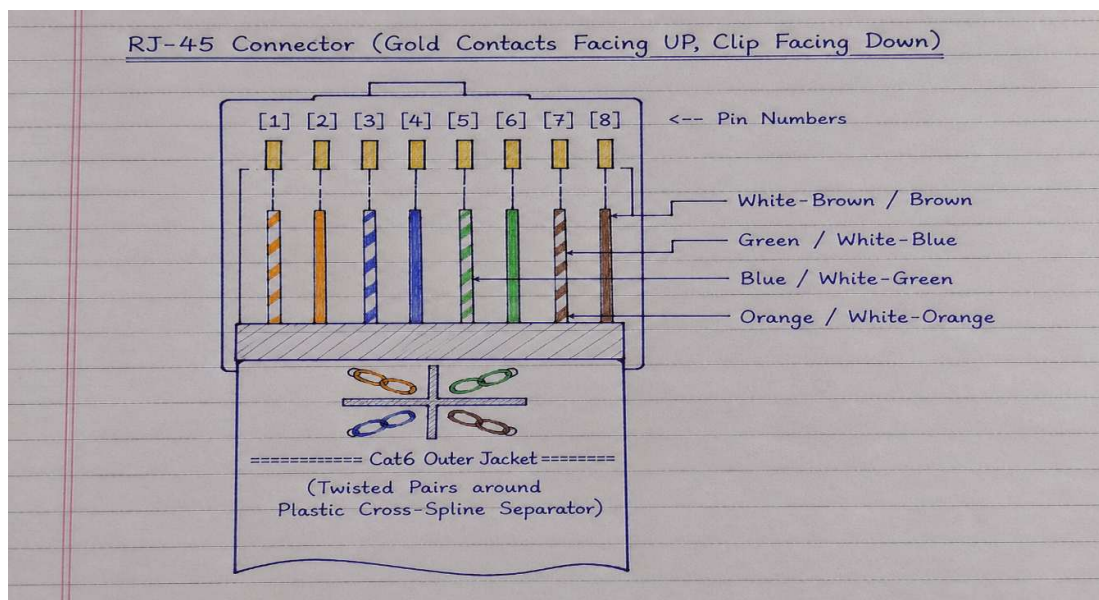
Objective: Diagram and specify the standard T568B color coding order for Cat6 Ethernet cable termination using an RJ-45 connector (Pins 1 to 8).

1. T568B Color Code Sequence (Pin 1 to Pin 8):

Pin Number	Wire Color Pattern	Function in 10/100/1000 Mbps
Pin 1	White-Orange	Transmit Data (+)
Pin 2	Orange	Transmit Data (-)
Pin 3	White-Green	Receive Data (+)
Pin 4	Blue	Unused (100M) / Data (1G)

Pin Number	Wire Color Pattern	Function in 10/100/1000 Mbps
Pin 5	White-Blue	Unused (100M) / Data (1G)
Pin 6	Green	Receive Data (-)
Pin 7	White-Brown	Unused (100M) / Data (1G)
Pin 8	Brown	Unused (100M) / Data (1G)

2. Text-Based Cat6 RJ-45 Connector Diagram:



Task 2:

Types of Network Cables, Devices, and Use Cases

Objective: Identify three common types of network cabling, map each to a real-world device/application, and explain the technical reasoning for its deployment.

- **Cat5e Cable (Category 5e Unshielded Twisted Pair)**
 - **Real-World Device: Zomato POS Terminal / Retail Billing Counter**
 - **Reasoning:** POS billing systems require simple, cost-effective, and reliable wired LAN connections to process transactions. Since data bandwidth requirements for text receipt processing are low (100 Mbps to 1 Gbps max), Cat5e is the most budget-friendly cable choice.
- **Cat6 Cable (Category 6 Gigabit Ethernet Cable)**

- **Real-World Device: High-Speed Home/Office Wi-Fi Router**
- **Reasoning:** Cat6 supports speed up to 1 Gbps (and 10 Gbps over short distances) with strict shielding to reduce crosstalk. It provides optimal bandwidth to support high-speed ISP fibers, multiple simultaneous 4K streams, and online gaming without lag.
- **Fiber Optic Cable (Single-Mode / Multi-Mode Fiber)**
- **Real-World Device: IPL Streaming Cloud Server / Data Center Core Switch**
- **Reasoning:** Fiber optic cables transmit data using light pulses over long distances with immunity to electromagnetic interference (EMI). They provide ultra-high bandwidth (10Gbps to 100Gbps+) needed to handle millions of concurrent live streaming users during IPL matches without latency spikes.

Task 3:

Step-by-Step RJ-45 Connector Crimping Process

Objective: Document the procedure for terminating an Ethernet cable with an RJ-45 connector using the T568B standard, along with key mistakes to prevent.

1. Step-by-Step Crimping Process:

1. Strip the Outer Cable Jacket:

Use a wire stripper to carefully remove about 1 inch (2.5 cm) of the outer PVC jacket from the Cat6 Ethernet cable, taking care not to nick or cut the internal copper conductors.

2. Untwist and Straighten Wires:

Untwist the 4 pairs of wires and separate them. Remove any internal plastic spline or core strain relief thread using wire cutters.

3. Arrange Wires in T568B Color Order:

Align the individual wires side-by-side according to the standard **T568B color pinout**:

White-Orange → Orange → White-Green → Blue → White-Blue → Green → White-Brown → Brown.

4. Trim Wires Evenly:

Flatten the arranged wires and use the crimping tool blade to trim them straight across, leaving approximately 0.5 inches (1.2 cm) of exposed wires.

5. Insert Wires into RJ-45 Connector:

Hold the RJ-45 connector with the plastic locking tab facing down (gold contacts facing up). Push all 8 wires firmly into the connector until each wire reaches the front tip of the plug and the outer jacket extends inside the connector past the locking crimp wedge.

6. Crimp the Connector:

Insert the prepared RJ-45 connector into the crimping tool slot and squeeze the handles firmly together until the tool clicks/locks to punch down the gold pins into the wires.

7. Test Connection:

Use a Cable Tester to verify pin-to-pin continuity across all 8 pins on both ends of the cable.

2. Two Common Mistakes to Avoid:

1. Incorrect Cable Jacket Depth (Exposed Wires Outside Plug):

Stripping too much outer jacket or not pushing the cable far enough into the RJ-45 plug leaves bare individual wires exposed outside the connector. This reduces strain relief and leads to physical wire damage or loose connections over time.

2. Incorrect Wire Sequence / Mixed T568A and T568B Standards:

Accidentally swapping adjacent wires (like Green and White-Green) or terminating one end in T568A and the other in T568B without intending to make a crossover cable will result in split pairs, high crosstalk, or complete link failure.

Task 4:

Gaming Zone & IPL Screening Event Setup Scenario

Objective: Select optimal network cables, audio/video cables, and connector types to deliver zero-latency online gaming and high-definition live IPL stream display.

1. Recommended Cable and Connector Selections:

For Gaming Consoles & Streaming Hardware (Data Connection):

Cable Type: Cat6 / Cat6a Unshielded/Shielded Twisted Pair Cable

Connector Type: RJ-45 (Registered Jack 45)

For Display Screens & TVs (Audio/Video Connection):

Cable Type: Ultra High Speed HDMI 2.1 Cable

Connector Type: Standard HDMI Type-A Connector

2. Justification & Explanation of Choices:

Cat6 / Cat6a Cable with RJ-45 Connectors (Networking):

High Bandwidth & Low Latency: Cat6 supports up to 10 Gbps (and Cat6a up to 10 Gbps) bandwidth with low crosstalk, ensuring ultra-low ping and lag-free online multiplayer gaming for consoles.

Reliability: Physical RJ-45 wired connections eliminate Wi-Fi interference and packet loss caused by overcrowded public wireless signals during the event.

HDMI 2.1 Cable with HDMI Type-A Connectors (Video Output):

4K/8K High Refresh Rate: HDMI 2.1 handles up to 48 Gbps uncompressed bandwidth, allowing 4K resolution at 120Hz/60Hz live streaming for IPL matches without video tear or audio lag.

Universal Compatibility: Standard HDMI connectors lock securely into modern gaming consoles (PS5, Xbox, PCs) and large venue projectors/LED screens.

Task 5:

Practical Observation of Network Cable in Use

Objective: Document a real-world network cable deployment, identifying the cable category, connector type, and interconnected devices.

1. Description of Setup:

- **Cable Type: Category 6 (Cat6) UTP Ethernet Patch Cable (Yellow Color)**
- **Connector Type: RJ-45 (Registered Jack 45) Translucent Plastic Connector**
- **Location / Scenario: Home Wi-Fi Router Setup (Behind the Router)**

2. Connected Devices & Function:

- **Devices Connected:** Connects the **Optical Network Terminal (ONT / Fiber Modem)** output port directly to the **Wi-Fi Router's WAN (Internet) Port**.
- **Purpose:** It transfers high-speed digital data packets from the Fiber Modem to the Wi-Fi Router, enabling wireless internet distribution across home/office smartphones, laptops, and smart TVs.